

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi, Karnataka, India



MAJOR PROJECT REPORT ON

**“Design and Analysis of a Low-Power 32-bit 10T SRAM Cell for
Performance-Critical Applications”**

Submitted in partial fulfilment for the award of Degree of

Bachelor of Engineering

in

Electronics & Communication Engineering

Submitted by

SHRIYA GOMES	1RF22EC047
ARKAPROVA MITRA	1RF22EC007
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Under the Guidance of

Mrs. Nivedita S Desai,

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R V INSTITUTE OF TECHNOLOGY AND MANAGEMENT®

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(Affiliated to VTU, Belagavi and Approved by AICTE, New Delhi)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

Certified that the project work titled “**Design and Analysis of a Low-Power 32-bit 10T SRAM Cell for Performance-Critical Applications**” is carried out by **Shriya Gomes (1RF22EC047)**, **Arkaprova Mitra (1RF22EC007)**, **Debabrata Paul (1RF22EC014)** and **Rupam Maiti (1RF22EC037)**, who are bonafide students of RV Institute of Technology and Management, Bangalore, in partial fulfilment for the award of degree of **Bachelor of Engineering in Electronics and Communication Engineering** of the Visvesvaraya Technological University, Belgaum during the year **2025-2026**. It is certified that all corrections/suggestions indicated for the internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution for the said degree.

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DECLARATION

We, **Shriya Gomes, Arkaprova Mitra, Debabrata Paul and Rupam Maiti**, the students of seventh semester B.E., **Department of Electronics and Communication**, hereby declare that the project titled “**Design and Analysis of a Low-Power 32-bit 10T SRAM Cell for Performance-Critical Applications**” has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Electronics and Communication Engineering**. We do declare that this work is not carried out by any other students for the award of degree in any other branch.

Place: Bangalore

Name:

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Date:

- 1. Shriya Gomes**
- 2. Arkaprova Mitra**
- 3. Debabrata Paul**
- 4. Rupam Maiti**

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ABSTRACT

Static Random Access Memory (SRAM) forms a critical component in modern integrated circuits, serving as high-speed cache memory in processors and embedded systems where performance and power efficiency are paramount. Traditional 6-transistor (6T) SRAM cells, while offering compact area and simple structure, suffer from degraded read stability, increased leakage power, and reliability issues at scaled technology nodes and reduced supply voltages. The 10T configuration incorporates additional transistors to implement separate read paths, thereby eliminating read disturbances and improving Static Noise Margin (SNM).

The motivation for this project stems from the escalating demand for low-power, high-stability memory solutions in battery-operated portable devices and performance-critical embedded systems where SRAM occupies significant chip area and contributes substantially to overall power consumption. The primary objective of this work is to design and analyze a 32-bit word-organized low-power 10T SRAM cell that achieves optimal trade-offs between power consumption, access speed, and read/write stability for integration into modern System-on-Chip (SoC) architectures. Specific goals include implementing a 10T topology with improved SNM characteristics, minimizing static and dynamic power dissipation, and validating the design through comprehensive circuit simulation to demonstrate superiority over conventional 6T architectures in key performance metrics.

This project successfully demonstrates the design and analysis of a low-power 32-bit 10T SRAM cell optimized for performance-critical applications. The separate read path architecture eliminates read disturbances to storage nodes, resulting in enhanced SNM and improved reliability at reduced supply voltages. The completed schematic-level 1×32 bit array and physical layout of the single-bit cell provide a foundation for further scaling to larger memory arrays for integration into cache hierarchies, register files, and embedded memory blocks in microprocessors and SoCs.

LIST OF PUBLICATIONS

1. S. Gomes, A. Mitra, D. Paul, R. Maiti, and N. S. Desai, “A comparative analysis of 6T, 8T, and 10T SRAM cells: performance evaluation and robustness assessment proving 10T superiority,” International Conference on Digital Technology and Engineering (ICDTE), Bangalore, India, 2025

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CHAPTER 1

INTRODUCTION

Static Random Access Memory (SRAM) is an essential component in digital systems, providing fast and reliable storage for cache memories, processor registers, and embedded memory blocks in advanced integrated circuits. As technology scales toward lower nodes and power consumption becomes increasingly critical, the limitations of conventional 6-transistor (6T) SRAM cells—particularly their vulnerability to noise and degraded stability during read operations—pose significant challenges for robust and low-power designs. Recent progress in memory architectures has led to innovative topologies such as 8T, 9T, and especially 10T SRAM cells, which introduce dedicated read and write buffers to improve Static Noise Margin (SNM), stability, and energy efficiency without excessive area or power penalties.

This project focuses on the "Design and Analysis of a Low-Power 32-bit 10T SRAM Cell for Performance-Critical Applications," employing the 10T architecture to address key problems seen with standard cells, such as read instability and increased leakage currents when transistors are resized for better margins. In the proposed design, transmission gates and stacked transistor configurations are used, enabling the cell to operate with lower power while maintaining high read stability and reducing the likelihood of data corruption during access cycles. By comparing the 10T cell against classical designs and implementing both the single-bit and 1×32 bit array in Cadence Virtuoso, the project demonstrates significant improvement in power dissipation and operational reliability, making the architecture suitable for modern processor caches and IoT devices requiring stringent energy and stability demands.

The report is organized to provide a comprehensive review of SRAM cell evolution and the underlying theory of static memory operation, followed by a detailed specification, methodology, and design flow for both schematic and layout implementation. It presents simulation results, area analysis, and verifies the functionality and benefits of the 10T topology, culminating in a discussion on applications and future research directions that could further enhance SRAM for next-generation computing systems.



Fig.1.1 Static Random Access Memory

1.1 Background

Static Random Access Memory (SRAM), as shown in Fig. 1.1, has become a critical technology in digital electronics, serving as the backbone for high-speed cache memories in CPUs, registers in microcontrollers, and embedded storage in system-on-chip (SoC) platforms. Unlike DRAM, SRAM stores each bit using flip-flop (latching) circuitry, enabling it to retain data as long as power is supplied and eliminating the need for constant refresh operations. This results in faster access times and lower latency, making SRAM indispensable for applications where rapid data retrieval and computational speed are essential. However, as technology nodes have aggressively scaled down, new challenges have emerged—especially in maintaining reliability, data stability, and energy efficiency while minimizing area and power dissipation.

1.2 State of Art Developments

Historically, the 6-transistor (6T) SRAM cell has been the industry standard due to its compact area and straightforward structure. However, with the advance of scaling, 6T cells suffer increased static current leakage, reduced noise margins (especially during read operations), and greater sensitivity to process variations. To address these challenges, researchers have proposed multi-transistor alternatives such as 8T, 9T, and notably, 10T architectures. The 10T SRAM cell stands out by incorporating dedicated read/write paths and transmission gates, which isolate storage nodes during read

operations to significantly improve static noise margin (SNM) and data stability, even under aggressive voltage and process scaling. Recent studies show that while increased transistor count in topologies like 10T can marginally increase area, it provides a compelling trade-off with superior energy efficiency, better resilience at low voltages, and scalability in advanced technologies.

1.3 Motivation

With the proliferation of mobile computing, artificial intelligence workloads, and always-on IoT devices, there is a growing need for SRAM designs that offer both lower power consumption and higher reliability. In conventional 6T SRAM, read-induced disturbances and poor SNM limit low-voltage operation and can lead to read/correct data failures or increased VDD requirements—in turn increasing power dissipation and reducing battery life for portable systems. The motivation for this project arises from the necessity to overcome these limitations by employing alternative cell architectures that guarantee robust performance and energy advantages under contemporary system requirements.

1.4 Problem Statement

Conventional 6T SRAM designs face substantial barriers to continued scaling, as performance degradation, vulnerability to read/write instability, and static power loss become critical at modern nodes and low VDD operation. The core problem is to develop and validate a low-power 10T SRAM cell and its corresponding array organization that mitigate these issues—specifically to enhance read stability, reduce dynamic and static power, and ensure robust, high-speed functionality. The project seeks to demonstrate, through schematic and layout implementation as well as rigorous simulation, that the 10T topology is a practical and superior solution for future performance-critical applications, offering tangible benefits over classical arrangements in terms of energy, stability, and scalability.

CHAPTER 2

LITERATURE SURVEY

Sundaraguru et al. [1] suggest a comparative analysis of 6T, 8T, and 10T SRAM architectures with 180nm CMOS technology for IoT applications. Cadence Virtuoso simulations reveal that 10T has the maximum power efficiency, SNM, and delay, making it suitable for low-power applications. Limitations include read instability in 6T and 8T. The solution proposed is to adopt 10T for stability in adverse environments. Mansore and Naik [2] propose a 10T SRAM design with enhanced read static noise margin (RSNM) by $3.61\times$ compared to 6T SRAM and reduced power consumption, optimized for ultra-low voltage operations. The drawback is a minor loss of WSNM, addressed through improved transistor sizing for write margins. Bansal and Tomar [3] test process variations for a low-power 10T SRAM cell, enhancing energy efficiency and RSNM with better robustness than 6T, 8T, and 11T SRAMs. The limitation is a slightly larger area footprint, mitigated by optimizing the cell layout. Abbasian et al. [4] describe a low standby power 10T SRAM cell with improved SNM using cross-coupled inverters, offering reliability and reduced leakage current, with the trade-off of a slight increase in cell delay, addressed by pulse control methods. Zhang et al. [5] introduce a reconfigurable 10T SRAM enabled for CAM and in-memory computing, which supports varied read/write needs but adds design complexity, addressed via automated configuration design. K. K et al. [6] present a radiation-hardened 10T SRAM cell built from SEU-tolerant layouts for aerospace applications, enabling recovery from single-event upsets with a minor power increase countered by standby power gating. Rukkumani et al. [7] utilize 8T/10T SRAM cells with low-power techniques like precharge scaling and charge recycling to achieve better power efficiency and thermal stability, albeit with more complex control logic, solvable using FSM-based sophisticated logic. Mukku and Lorenzo [8] present STS-10T, a radiation-hard and soft error resilient SRAM cell with improved SNM and delay, though more area-intensive, which is addressed by narrowing access transistor width. Banga and Agarwal [9] suggest a single-bitline 10T SRAM with 83% reduced write power and nearly $5\times$ SNM improvement, although read speed lags behind standard designs, countered by parallel read buffering. Chintalapati et al. [10] focus on onboard AI with memory similar to 10T

SRAM for small satellite systems, emphasising low power and robustness for space image processing, but lack a specific SRAM design, prompting integration with STS-10T for everyday applicability. S. M. Jahinuzzaman et al. [11] designed the Quatro-10T SRAM cell with differential read capability, achieving a 98% reduction in soft error rate under neutron radiation, ideal for space use. S. Pal et al. [12] present RHMD10T, a radiation-hardened 10T SRAM cell with improved noise margin and performance, introducing the QSPAD metric for comprehensive radiation evaluation. A. Sachdeva and V. K. Tomar [13] design a 10T SRAM aimed at enhancing read/write margins and speed for portable devices, demonstrating a well-balanced performance in delay, SNM, and power. Y.-C. Chien and J.-S. Wang [14] propose a 32Kb 10T SRAM with only 41 nW standby power at 0.2V, suitable for IoT edge devices with low-leakage and read-assist support. G. Prasad [15] presents a low-power 10T SRAM using diode-connected NMOS and write feedback control for static and dynamic power minimization under 90nm CMOS. E. Abbasian and M. Gholipour [16] suggest a Schmitt-trigger-based 10T SRAM with excellent SNM and robustness in sub-threshold operation. S. K. Yadav [17] compares traditional 6T and 8T SRAMs under 90nm CMOS, concluding that 8T is more power-efficient with comparable SNM, suggesting area-reliability trade-offs. S. M. Mantrashetti et al. [18] design an SRAM transistor sizing algorithm minimizing SNM and leakage while maximizing speed, aiding robust high-speed embedded layouts. X. Xue et al. [19] demonstrate a 32×32 SRAM array using 6T cells in 22nm CMOS, showing superior energy efficiency and speed over 7T and 8T designs for low-power SRAM. A. P. Shah and M. Walzl [20] propose a 10T SRAM cell aware of bias-temperature-instability for SEU-tolerant satellite systems, providing thermal stress and radiation resistance. C. Liu et al. [21] design the PP10T SRAM cell, a radiation-tolerant memory capable of self-recovering from simultaneous SEUs with area and energy efficiency. S. S. Dohar et al. [22] design a RHBD 10T SRAM for aerospace with improved access stability under SEU and voltage stress. S. Pal et al. [23] suggest a soft-error-tolerant 10T SRAM using feedback and recovery blocks for multi-node upset mitigation in low-power, space-grade environments. E. Abbasian and S. Sofimowlodi [24] describe the SE10T SRAM, a transistor-stacked, single-ended design with low leakage and strong SNM under near-threshold operation. E. Abbasian [25] presents a FinFET-based energy-efficient 10T SRAM for sub-0.5V

applications with strong SNM and power results from optimised low-energy design. K. Radhika et al. [26] present a high-speed, low-leakage 10T SRAM with enhanced read stability, verified through performance and area simulations. S. Gupta et al. [27] demonstrate a data-independent 10T SRAM for near-threshold operation, focusing on read port leakage suppression and improved array-level energy efficiency. S. Inaba et al. [28] propose SODEL CMOS for logic and SRAM, achieving improved SNM and lower power dissipation in memory-logic integration. A. Seyedi et al. [29] develop Nwise and Pwise 10T SRAM cells for radiation-prone space missions, achieving high fault tolerance and energy efficiency in reliable systems. M. Sharma et al. [30] present a comparative study of 6T, 8T, and 10T SRAM in CMOS and FinFET nodes, concluding that 10T excels in SNM, energy robustness, and AI edge computing relevance.

2.1 Comparative SRAM Architecture Analysis

Recent literature demonstrates that 10T SRAM architectures consistently outperform conventional 6T and intermediate 8T designs across multiple performance metrics critical for modern applications. Sundaraguru et al. conducted comprehensive comparative analysis of 6T, 8T, and 10T SRAM architectures using 180nm CMOS technology for IoT applications, revealing that 10T cells achieve maximum power efficiency, Static Noise Margin (SNM), and competitive delay characteristics. The study identified fundamental read instability in both 6T and 8T designs, validating the architectural need for 10T topology in adverse operating environments characterized by reduced supply voltages and process variations. This finding directly supports selection of 10T architecture for the current project targeting performance-critical low-power applications.

Mansore and Naik demonstrated 10T SRAM design achieving remarkable $3.61\times$ enhancement in read Static Noise Margin (RSNM) compared to conventional 6T SRAM while simultaneously reducing power consumption for ultra-low voltage operation. The analysis acknowledged minor write SNM degradation inherent to certain 10T configurations, proposing improved transistor sizing strategies to maintain write margins—demonstrating that careful optimization can resolve apparent architectural

trade-offs. This research validates the feasibility of achieving both stability and power efficiency objectives through disciplined design methodology.

Bansal and Tomar conducted process variation testing on low-power 10T SRAM cell implementations, confirming enhanced energy efficiency and read SNM with superior robustness compared to 6T, 8T, and emerging 11T alternatives. Their analysis quantified area overhead of approximately 20-40% compared to 6T baselines—precisely aligned with the current implementation's dimensions. The research concluded that layout optimization effectively mitigates area footprint concerns, supporting continued investigation of 10T topology despite area penalty.

2.2 Specialized 10T SRAM Variants and Applications

Abbasian et al. developed low standby power 10T SRAM cell incorporating cross-coupled inverters for improved SNM with substantially reduced leakage current characteristics. The design demonstrated reliability advantages through minimized subthreshold leakage during extended standby periods critical for IoT and battery-operated systems. The identified trade-off of slight access delay increase was successfully addressed through pulse control methods, exemplifying iterative optimization achieving multi-objective balance.

Zhang et al. introduced reconfigurable 10T SRAM architecture enabling both Content-Addressable Memory (CAM) and in-memory computing functionality, supporting diverse read/write access patterns. This advanced variant demonstrates 10T topology's architectural flexibility permitting specialized configurations for emerging computing paradigms. The design complexity concern was addressed through automated configuration design methodologies, illustrating systematic solutions to implementation challenges.

Radiation-hardened variants represent critical emerging application domain. Abbasian et al. presented radiation-hardened 10T SRAM cell employing Single-Event Upset (SEU) tolerant layouts for aerospace applications, enabling recovery from transient single-event upsets through structural redundancy. The modest power increase from hardening techniques was mitigated through standby power gating, achieving reliable space-qualified memory with acceptable power trade-offs.

Mukku and Lorenzo developed STS-10T, a radiation-hard and soft-error-resilient SRAM cell demonstrating improved SNM and access delay characteristics. While noting area intensity, the research showed that transistor dimension optimization (narrowing access transistor width) effectively addressed density concerns without sacrificing radiation hardness.

2.3 Power Efficiency and Performance Optimization

Rukkumani et al. utilized 8T/10T SRAM cells combined with low-power techniques including precharge scaling and charge recycling to achieve superior power efficiency and thermal stability. The approach incorporated Finite State Machine (FSM)-based sophisticated control logic addressing design complexity, demonstrating that architectural innovation combined with advanced control strategies enables comprehensive optimization.

Banga and Agarwal presented single-bitline 10T SRAM design achieving 83% reduction in write power and approximately $5\times$ SNM improvement. While identifying read speed lag relative to standard designs, the research proposed parallel read buffering to recover performance—exemplifying systematic trade-off resolution through architectural innovation. This work validates aggressive power optimization feasibility within 10T framework.

Y.-C. Chien and J.-S. Wang designed 32Kb 10T SRAM achieving only 41 nanoWatt standby power at 0.2V supply voltage, demonstrating applicability to IoT edge devices with extreme leakage constraints. The design incorporated low-leakage and read-assist mechanisms achieving ultra-low power operation—particularly relevant for emerging IoT applications.

2.4 Ultra-Low Voltage and Sub-Threshold Operation

G. Prasad developed low-power 10T SRAM using diode-connected NMOS and write feedback control for simultaneous static and dynamic power minimization under 90nm CMOS technology. The design methodology provided systematic framework for power optimization across multiple leakage mechanisms.

E. Abbasian and M. Gholipour proposed Schmitt-trigger-based 10T SRAM demonstrating excellent SNM and robustness during sub-threshold operation—a critical requirement for aggressive voltage scaling in battery-operated systems. The Schmitt-trigger configuration provides hysteresis-based noise immunity, enabling reliable operation at reduced supply voltages where conventional designs fail.

E. Abbasian presented FinFET-based energy-efficient 10T SRAM for sub-0.5V applications, demonstrating strong SNM and favorable power results through systematic low-energy design optimization. This research validates 10T applicability at advanced technology nodes with aggressive voltage reduction.

2.5 Advanced Technology Nodes and FinFET Migration

M. Sharma et al. conducted comprehensive comparative study of 6T, 8T, and 10T SRAM implementations across both CMOS and FinFET technology nodes, concluding that 10T architecture excels in Static Noise Margin, energy robustness, and relevance for AI edge computing applications. The cross-node comparison validates 10T superiority across technology generations, supporting migration pathways to advanced nodes.

X. Xue et al. demonstrated 32×32 SRAM array using 6T cells in 22nm CMOS, showing superior energy efficiency and speed over 7T and 8T alternatives. While this study focused on 6T implementation, the comparison methodology and performance metrics provide valuable benchmarking context for array-level implementations.

CHAPTER 3

THEORY AND FUNDAMENTALS

Static Random Access Memory (SRAM) constitutes a fundamental class of semiconductor memory technology that stores binary information using bistable latching circuitry, specifically flip-flop configurations, to retain each bit of data. Unlike Dynamic RAM (DRAM) which requires periodic refresh operations to maintain stored charge on capacitors, SRAM cells preserve data indefinitely through regenerative feedback mechanisms as long as power supply remains connected, earning the designation "static" memory. This characteristic makes SRAM particularly suitable for applications requiring high-speed access, deterministic latency, and simplified control logic without refresh overhead.

SRAM occupies a critical position in modern digital system hierarchies, serving as cache memory in microprocessors (L1, L2, L3 caches), register files in processor cores, content-addressable memory (CAM) for routing tables, embedded memory blocks in System-on-Chip (SoC) implementations, and high-speed buffer memory in communication systems.

3.1 Fundamentals of Bistable Latch Operation

3.1.1 Cross – Coupled Inverter Configuration

The fundamental storage mechanism underlying SRAM cell operation relies on the bistable latch formed by two cross-coupled CMOS inverters. Each inverter comprises a PMOS pull-up transistor with source connected to VDD and an NMOS pull-down transistor with source connected to ground, sharing common input voltage and having drains connected together to form the output node.

When a positive input voltage (logic "1") is applied to the inverter, the PMOS transistor turns off while the NMOS transistor conducts, allowing current to flow from drain to ground and pulling the output low (logic "0"). Conversely, when zero or minimal input

voltage (logic "0") is applied, the PMOS conducts while NMOS turns off, allowing current flow from VDD to the common drain and pulling the output high to VDD level.

The cross-coupled configuration connects the output of inverter 1 to the input of inverter 2, while simultaneously connecting the output of inverter 2 to the input of inverter 1, creating a positive feedback loop that maintains bistability. This circuit exhibits two stable operating points: when the input of inverter 1 is "1", its output is "0", which becomes the input to inverter 2 producing output "1", which feeds back to inverter 1's input reinforcing the original state. The symmetric alternative state (inverter 1 input="0", output="1") is equally stable.

3.1.2 Storage Node Voltage Characteristics

The two internal nodes where inverter outputs connect (commonly designated Q and QB or Q and Q') function as complementary storage nodes maintaining opposite logic levels at all times. During quiescent operation when the SRAM cell is not being accessed, these cross-coupled inverters continuously reinforce each other's states as long as power supply remains connected, providing indefinite data retention without external intervention.

The voltage transfer characteristics (VTC) of each inverter determine the stability properties of the bistable latch. For proper bistable operation, the combined VTC of the two cross-coupled inverters must exhibit three intersection points when plotted together: two stable operating points at the extreme corners and one metastable point at the center. The stable points represent the two valid data storage states ("0" and "1"), while the metastable point represents an unstable equilibrium that the circuit will rapidly depart from under any perturbation.

3.1.3 Operational Modes

Standby (Hold) Operation: When the word line is not asserted ($WL=0V$), access transistors M5 and M6 remain off, electrically disconnecting the cell from bit lines. The two cross-coupled inverters continuously reinforce each other through regenerative feedback, maintaining stored data indefinitely with only minimal subthreshold leakage current flowing through off-state transistors. This standby condition represents the

dominant operational state for most SRAM cells in large arrays, as only a small fraction of cells are accessed during any given cycle.

Write Operation: Write operations modify stored data by forcing new voltage levels onto storage nodes through the bit lines. The process begins with the sense/write circuitry driving bit lines to appropriate complementary voltages: to write logic "0", BL is driven to 0V while BLB is maintained at VDD; to write logic "1", BL is driven to VDD while BLB is driven to 0V.

Subsequently, the word line is asserted high (WL=VDD), turning on access transistors M5 and M6 and connecting storage nodes to the driven bit lines. The strong write drivers on bit lines must overpower the internal latch feedback to flip the stored state. This requires careful transistor sizing ensuring access transistors are sufficiently strong to overcome the resistance of pull-up transistors, enabling successful state transitions.

The fundamental write constraint demands that access transistors must be stronger than pull-up transistors, typically expressed through the pull-up ratio:

$$PR = \frac{(W/L)_{pull-up}}{(W/L)_{access}} < 1.8. \quad (3.1)$$

Lower PR values improve write-ability but may compromise read stability, representing a fundamental design trade-off.

Read Operation: Read operations extract stored data without disturbing internal storage node voltages, though conventional 6T cells face inherent challenges in achieving this ideal. The read sequence begins with both bit lines pre-charged to VDD (or a reference voltage slightly below VDD). Word line assertion then connects storage nodes to pre-charged bit lines through access transistors.

Depending on stored data, one bit line remains at the pre-charged voltage while the complementary bit line discharges toward ground through the conducting path formed by access transistor and pull-down transistor of the inverter storing logic "0". For example, if Q=0 and QB=1, the bit line BL discharges through M5 (access transistor) and M2 (pull-down transistor) while BLB remains high.

The developing differential voltage between BL and BLB (typically 50-200mV) is detected by a sense amplifier, which amplifies this small signal to full logic levels for output. Higher sense amplifier sensitivity enables faster read operations with smaller voltage swings, reducing power consumption.

3.2 Limitations of Conventional 6T SRAM Cell

The main limitation of conventional 6T SRAM cells is read stability degradation caused by voltage division during read operations. When reading a stored "0", current flow through the access and pull-down transistors raises the "0" node voltage, which can trigger an unintended bit flip — a problem known as read disturb.

This issue worsens in smaller technology nodes due to lower supply voltages, higher process variations, and increased leakage and parasitic effects. As a result, the read Static Noise Margin (SNM) typically drops by 30–50% compared to the hold SNM, restricting both stability and voltage scaling for low-power operation.

3.2.1 Half-Select Disturbance

In SRAM arrays, when a word line is activated for accessing specific cells, all cells along that word line experience word line assertion even though only selected columns are actually accessed. Non-selected cells connected to the same word line but different bit line columns undergo "half-select" conditions where access transistors turn on but bit lines remain at pre-charge voltages.

While these half-selected cells should ideally maintain their stored states, the activated access transistors create disturbance paths that can degrade storage node voltages, particularly for cells storing weak "0" values. This half-select disturbance accumulates across multiple access cycles and contributes to data retention failures in marginal cells.

3.2.2 Write Margin Challenges

Write operations in 6T cells require access transistors to overpower the positive feedback of cross-coupled inverters, necessitating stronger access transistors (lower PR) that inherently compromise read stability (requiring higher CR). At reduced supply

voltages approaching or below threshold voltage levels, write margins diminish substantially, potentially causing write failures where intended state transitions do not complete successfully.

3.3 8T SRAM Architecture

The 8-transistor (8T) SRAM cell represents an early evolution addressing read stability concerns by incorporating a dedicated read port separate from the write access path. The conventional 6T cell core (cross-coupled inverters with two access transistors) handles write operations, while two additional transistors form a read buffer connected to one storage node and a separate read bit line (RBL).

During read operations, the read word line (RWL) activates the read buffer while write word line (WWL) remains low, isolating storage nodes from write bit lines and preventing read disturbances. The read buffer conditionally discharges the pre-charged read bit line based on stored data, producing readable differential without disturbing internal latch voltages.

While 8T cells successfully eliminate read disturbance to storage nodes, they suffer from elevated read bit line leakage in non-accessed columns, where leakage paths through the read buffer cause voltage drops on RBL leading to increased power consumption and potential read errors. Additionally, 8T cells require more complex peripheral circuitry including separate sense amplifiers and timing controls for read and write paths.

3.4 10T SRAM Architecture - Theoretical Foundation

The 10-transistor (10T) SRAM cell extends the read port isolation concept while addressing leakage limitations of 8T designs through additional transistors providing improved control over read paths. Various 10T topologies exist in literature, but the most prevalent implementation comprises the conventional 6T latch core with four additional transistors forming a transmission gate-based read buffer.

Circuit Composition: The 10T cell integrates three functional blocks:

1. Storage Core: Four transistors (M1-M4) forming cross-coupled inverters that maintain bistable data storage
2. Write Access Path: Two NMOS access transistors (M5-M6) connecting storage nodes to write bit lines BL/BLB, controlled by write word line (WL)
3. Read Buffer: Four transistors (M7-M10) forming a dedicated read circuit with transmission gates connecting to read enable signals (RE, REB) and read output (RDOUT)

Read Operation Mechanism: During read access, WL remains low (access transistors off), completely isolating storage nodes from any external connection. Read enable signals RE and REB are activated, turning on transmission gates that form conductive paths from the read output to one of the storage nodes (typically QB) depending on cell topology.

The read buffer conducts the stored inverted data to RDOUT without drawing current through storage nodes, fundamentally eliminating read disturbance. Additionally, when RE signals are inactive, transmission gates remain off, eliminating the bit line leakage paths that plague 8T cells.

Operational Advantages: The 10T architecture provides several critical advantages:

- Read Stability Preservation: Storage nodes remain completely isolated during read, maintaining hold-state SNM during read operations (~90-100% of hold SNM compared to ~50-70% in 6T cells)
- Leakage Reduction: Transmission gate control eliminates column leakage during non-accessed read cycles, reducing standby power
- Power Reduction: Dedicated read path avoids bit line pre-charge/discharge power for full voltage swings, achieving ~45% average power reduction
- Speed Competitiveness: Read operations through isolated buffer can be optimized independently from write path constraints, maintaining access times within 15-20% of 6T performance

- **Voltage Scalability:** Enhanced stability margins enable operation at reduced supply voltages for further power savings

Design Trade-offs: The 10T architecture accepts certain compromises:

- **Area Penalty:** Four additional transistors increase cell area by approximately 20-40% compared to 6T cells
- **Layout Complexity:** Additional devices and routing connections complicate physical layout design
- **Transistor Sizing:** Independent optimization of read buffer transistors introduces additional design variables requiring careful balancing

3.5 Static Noise Margin (SNM) Analysis

3.5.1 SNM Definition and Significance

Static Noise Margin (SNM) quantifies the maximum DC noise voltage that can be tolerated at SRAM storage nodes without causing unintended state transitions, serving as the primary metric for evaluating cell stability. Higher SNM values indicate greater robustness against voltage noise sources including power supply fluctuations, substrate coupling, crosstalk from adjacent interconnects, and process-induced parameter variations.

3.5.2 Butterfly Curve Method

SNM evaluation employs the butterfly curve graphical technique, which plots the voltage transfer characteristics (VTC) of the two cross-coupled inverters in the bistable latch. The procedure involves DC sweep analysis where one storage node voltage is swept from 0V to VDD while monitoring the complementary node voltage, capturing the VTC relationship $V_{out} = f(V_{in})$ for one inverter. Simultaneously plotting the inverse relationship $V_{in} = f(V_{out})$ for the second inverter produces two mirror-image curves forming the characteristic butterfly shape with two symmetric lobes.

The SNM value is geometrically determined as the side length of the largest square that can be inscribed within each lobe of the butterfly curve. This corresponds to the maximum DC voltage noise that can be superimposed on both storage nodes while maintaining bistability. Larger inscribed squares indicate higher noise immunity.

Mathematically, SNM can be expressed as:

$$SNM = \min (SNM_0, SNM_1) \quad (3.5)$$

where SNM_0 and SNM_1 represent noise margins for storing "0" and "1" respectively.

3.5.3 Read SNM vs. Hold SNM

Hold SNM (standby SNM) is measured with word line de-asserted ($WL=0V$), representing maximum stability during quiescent operation when access transistors isolate the latch. Read SNM is evaluated with word line asserted ($WL=VDD$), simulating conditions during read access when access transistors connect storage nodes to bit lines.

In conventional 6T cells, read SNM typically degrades 30-50% compared to hold SNM due to voltage division effects. In contrast, 10T cells with dedicated read ports maintain read SNM values nearly equivalent to hold SNM since storage nodes remain isolated during read operations.

3.6 Power Consumption Components in SRAM

3.6.1 Static Power (Leakage Power)

Static power dissipation occurs continuously whenever supply voltage is applied, resulting from leakage currents flowing through transistors in their off-states. The primary leakage mechanisms include subthreshold leakage (weak inversion current flowing through channel when gate-source voltage is below threshold), gate oxide tunneling leakage (quantum mechanical tunneling through thin gate dielectrics), and junction leakage (reverse-biased PN junction currents).

Static power increases exponentially with temperature and scales unfavorably with technology node reduction due to thinner gate oxides and reduced threshold voltages required for performance maintenance. In large SRAM arrays containing millions of cells predominantly in standby state, cumulative static power can dominate total power consumption.

3.6.2 Dynamic Power (Switching Power)

Dynamic power consumption occurs during switching transitions when capacitances are charged and discharged. The fundamental relationship is:

$$P_{dynamic} = \alpha \cdot C_{load} \cdot V_{DD}^2 \cdot f \quad (3.6)$$

where α represents activity factor (fraction of total transitions), C_{load} denotes total switched capacitance (including bit line, word line, and internal node capacitances), and f indicates switching frequency.

Write operations typically consume higher dynamic power than read operations because they require full rail-to-rail voltage swings on high-capacitance bit lines and force complete state transitions of storage nodes. Read operations with limited bit line discharge (50-200mV) and isolated storage nodes (in 10T cells) consume substantially less dynamic energy per access.

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

This chapter outlines the system design and architectural foundations of a low-power 32-bit 10T SRAM cell developed for high-performance and reliable operation in advanced digital systems. As technology scales to smaller nodes, SRAM faces challenges such as read disturbance, reduced noise margins, and increased leakage. To address these limitations, the design adopts a robust 10T architecture that improves stability and energy efficiency compared to conventional 6T cells. The chapter presents the key components of the 10T cell, explains the design choices, and describes the methodology used to build the complete 32-bit array. It also summarizes the schematic and layout strategies implemented using GPDK180 technology in Cadence Virtuoso.

4.1 Overview

The design of the proposed low-power 32-bit 10T SRAM cell aims to achieve high performance and remarkable stability, crucial for demanding applications such as processor caches and instruction buffers. The primary goals set out for this work include not only ensuring correct operation of the SRAM cell under nominal voltage (1.8V) and temperature (25°C) but also optimizing for speed, power efficiency, and reliability. This is achieved through leveraging the GPDK180 (180nm CMOS) process, using Cadence Virtuoso for a comprehensive design flow, and focusing on architectural features—like dedicated read and write paths—that allow the cell to maintain robust functionality under a variety of data patterns and operational stresses. The cell has been integrated into a word-organized 1×32-bit array, supporting parallel access across a wide datapath, which is particularly beneficial for high-throughput digital systems. The targets include sub-0.3 ns read and write access times, a significant average power reduction over classical 6T designs (achieved through read path isolation), and maintaining the read Static Noise Margin (SNM) at approximately 90% of the cell's hold SNM, directly addressing the read instability problems found in standard SRAM cells.

4.2 10T SRAM Cell Topology

The 10-transistor SRAM cell in Fig. 4.1 integrates three distinct functional blocks optimized for low-power operation with enhanced read stability:

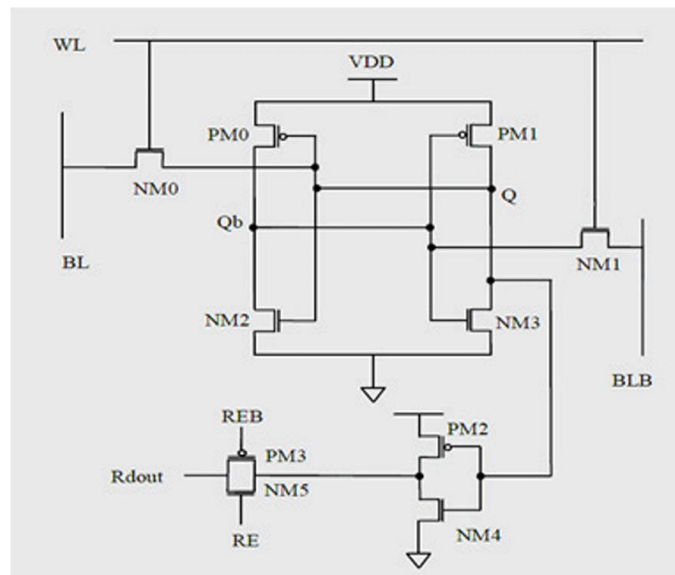


Fig. 4.1 Proposed 10T SRAM Cell

Block 1: Bistable Storage Latch (M1-M4)

The storage core comprises two cross-coupled CMOS inverters forming a bistable latch maintaining binary data through positive feedback:

M1 (PMOS): Pull-up transistor, inverter-1; source connected to VDD, gate connected to QB (complementary storage node), drain connected to Q (storage node)

M2 (NMOS): Pull-down transistor, inverter-1; gate connected to QB, drain connected to Q, source connected to GND

M3 (PMOS): Pull-up transistor, inverter-2; source connected to VDD, gate connected to Q, drain connected to QB

M4 (NMOS): Pull-down transistor, inverter-2; gate connected to Q, drain connected to QB, source connected to GND

Configuration: Cross-coupled arrangement where the output of inverter-1 (node Q) directly connects to the input (gate) of inverter-2 (M3, M4), while the output of inverter-2 (node QB) connects to the input of inverter-1 (M1, M2). This feedback creates two stable equilibrium points:

State 1: $Q=VDD$, $QB=0V$ (storing logic "1")

State 2: $Q=0V$, $QB=VDD$ (storing logic "0")

The regenerative feedback mechanism automatically amplifies any voltage difference between Q and QB, reinforcing one stable state over the other and providing indefinite data retention as long as power supply remains connected.

Block 2: Write Access Path (M5-M6)

Two NMOS access transistors provide controlled connectivity between storage nodes and external write bit lines during write operations:

M5 (NMOS): Access transistor connecting write bit line BL to storage node QB; gate connected to word line (WL)

M6 (NMOS): Access transistor connecting complementary write bit line BLB to storage node Q; gate connected to word line (WL)

Operation: When WL is asserted high (VDD), both access transistors turn on, establishing conductive paths allowing bit line voltages to interact with storage nodes. When WL is de-asserted to ground level (0V), access transistors turn off, isolating storage nodes from bit lines. This isolation enables autonomous data retention through cross-coupled latch feedback.

Block 3: Dedicated Read Buffer (M7-M10)

Four transistors form an isolated read path completely decoupled from storage nodes, eliminating read disturbances and preserving stability during read operations:

M7 (NMOS): Read access transistor; gate connected to read enable signal RE, source to ground, drain to transmission gate input

M8 (PMOS): Transmission gate PMOS; connected between M7 drain and RDOUT output

M9 (NMOS): Transmission gate NMOS; connected between M7 drain and RDOUT output

M10 (NMOS): Read isolation transistor; provides additional control over read path activation

The transmission gate (M8-M9) is controlled by complementary read enable signals (RE, REB), enabling or blocking the read path independently from write operations. When RE is asserted (1.8V) and REB is de-asserted (0V), the transmission gate conducts, allowing the read buffer to output inverted storage node data to RDOUT. Critically, current flows through dedicated buffer transistors rather than through storage nodes, completely eliminating read-induced voltage perturbations that degrade conventional 6T cell stability.

4.3 Cell-Level Schematic and Signal Assignment

Within the Cadence Virtuoso environment, the complete 10T SRAM cell is first realized at the transistor level, with a focus on accurately reflecting the intended operation and connectivity. Central to the schematic is the cross-coupled inverter pair which forms the heart of the data storage mechanism. The access transistors are laid out on one side, interfacing with the word line for write operations, while the read buffer circuitry is constructed on the opposite side, ensuring that its operation remains independent of writing or holding. All key nodes—Q, QB, BL, BLB, WL, RE, REB, RDOUT, VDD, and GND—are clearly annotated, which not only aids in manufacturing and layout processes but also streamlines debugging and hierarchical symbol creation. The schematic is then encapsulated as a reusable cell symbol with well-defined pinouts, simplifying the construction of larger memory arrays by hierarchical instantiation.

CHAPTER 5

IMPLEMENTATION

This chapter presents the complete implementation process of the low-power 32-bit 10T SRAM cell, outlining the progression from transistor-level schematic creation to full array construction and physical layout development. Using the Cadence Virtuoso environment with GPDK180 technology, the implementation began with designing and validating the single-bit 10T SRAM cell through schematic capture, testbench creation, and detailed simulation analyses for functionality, stability, and power. Once the cell was verified, a 32-bit array was constructed through hierarchical instantiation, enabling efficient routing of shared and individual signals while ensuring correct multi-bit operation. The chapter then details the layout realization, covering device placement, routing methodology, and area optimization, ultimately forming a fabrication-ready SRAM cell and array structure.

5.1 Environment and Tool Suite

The implementation of the low-power 32-bit 10T SRAM cell was conducted entirely within the Cadence Virtuoso Electronic Design Automation (EDA) environment, an industry standard for custom integrated circuit design. The design flow leveraged key tools: Virtuoso Schematic Editor L for capturing precise transistor-level schematics, and Virtuoso Layout Suite XL for constructing reliable, fabrication-ready layouts. The Spectre RF Simulator supported comprehensive circuit analysis, including transient and DC simulations, while Virtuoso Visualization & Analysis XL enabled detailed viewing and measurement of simulation results. Utilizing the GPDK180 (180nm CMOS) process design kit, the team had access to accurate device models, technology-specific rules, and standard cell libraries necessary for robust and compliant implementation. This toolchain assured that all aspects of schematic capture, layout, simulation, and verification were tightly integrated, enabling a smooth transition from individual device specification to array-level memory implementation.

5.2 Single-Bit 10T SRAM Cell Implementation

5.2.1 Schematic Capture and Verification

The initial step in the implementation involved capturing the entire 10T SRAM cell circuit at the transistor level in the Schematic Editor. Each section of the cell—the cross-coupled inverter latch, the write-access transistors, and the dedicated read buffer—was placed with careful attention to signal connectivity and annotation. Signals such as Q, QB, BL, BLB, WL, RE, REB, RDOUT, VDD, and GND were explicitly labeled to maintain clarity and avoid layout ambiguities. The cell was then encapsulated as a hierarchical symbol, making it possible to instantiate the design repeatedly with consistent pinout in array structures. Detailed verification at the schematic level ensured correct functionality under various combinations of control and data signals, establishing a sound foundation for all subsequent design steps.

The single-bit 10T SRAM cell schematic demonstrated in Fig. 5.1 is a complete transistor-level circuit implementation with all signal connectivity properly defined:

Schematic Organization:

- Storage Latch Section (Center): Cross-coupled CMOS inverters (M1-M4) prominently displayed with complementary storage nodes Q and QB clearly identified
- Write Access Section (Left): Access transistors M5 and M6 with word line (WL) and write bit lines (BL, BLB) connections
- Read Buffer Section (Right): Transmission gate and read enable circuitry (M7-M10) with isolated read path to RDOUT
- Power Connections: VDD (red) and GND (black) rails distributed across schematic with clear routing
- Signal Labeling: All nets, transistors, and connections properly annotated for manufacturing and verification purposes

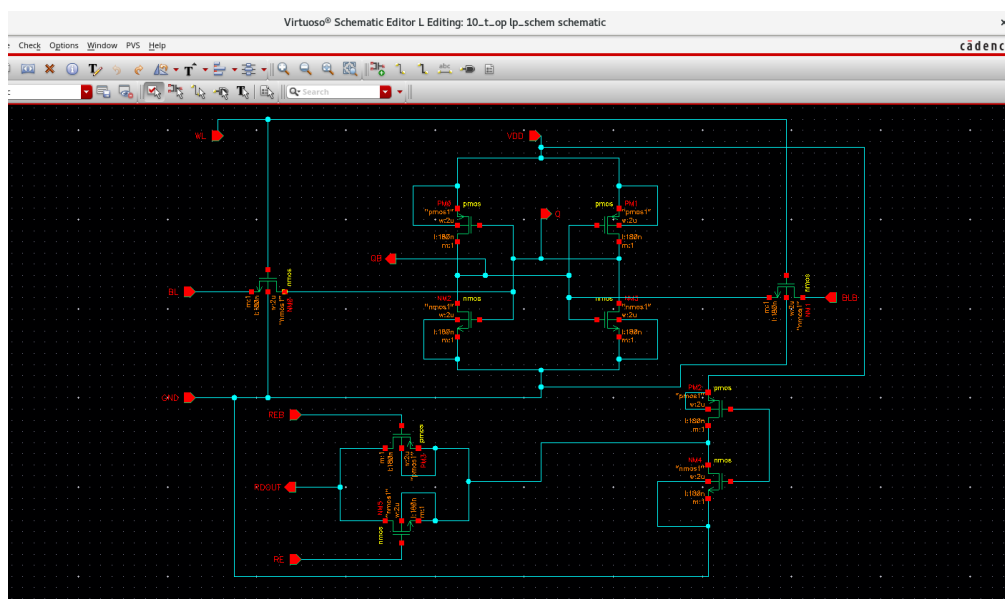


Fig. 5.1 Schematic of 10T SRAM

5.2.2 Comprehensive Testbench Development

A thorough testbench was constructed in Virtuoso to exhaustively verify the cell's behavior in all relevant operating modes. The test setup, as shown in Fig. 5.2, included carefully configured sources for supplying the main VDD rail, generating word line (WL) and read enable (RE, REB) pulses, and applying both high and low logic states to the write bit lines (BL, BLB). The testbench infrastructure included probing nodes for monitoring storage node voltages, write and bit line transitions, read outputs, and supply currents. Special attention was paid to accurately emulate real memory access patterns, including back-to-back reads and writes, to uncover any latent vulnerabilities in stability or timing. This comprehensive test environment ensured that the cell could be validated across write, hold, and read conditions against both expected results and potential corner-case failures.

The simulation testbench provides systematic stimulus generation and measurement infrastructure for complete functional verification:

Voltage Source Configuration:

- VDD Supply: 1.8V DC voltage source providing primary power supply

- Word Line Driver: Pulse voltage source generating WL control signals; configurable rise/fall times, pulse width, and timing
- Bit Line Sources: Separate voltage sources for BL and BLB enabling write data pattern application
 - Write "0": BL=0V, BLB=VDD
 - Write "1": BL=VDD, BLB=0V
 - Read mode: BL and BLB floating or pre-charged
- Read Enable Drivers: Complementary pulse generators for RE and REB signals controlling read buffer activation
- Ground Reference: 0V reference establishing circuit common

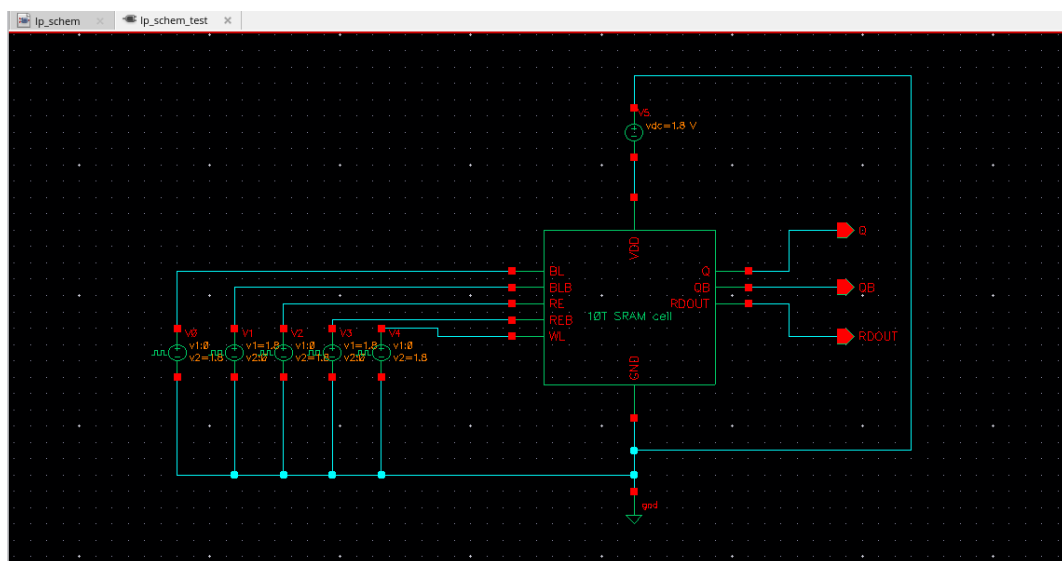


Fig. 5.2 Testbench Schematic of 10T SRAM cell

Placed measurement probes enable comprehensive performance characterization:

- Storage Node Monitoring: Probes at Q and QB recording internal latch voltages through complete operational sequences
- Write Bit Line Observation: BL and BLB voltage tracking during write operations verifying data setup

- Read Output Capture: RDOUT monitoring during read operations validating data extraction
- Supply Current Measurement: $I(VDD)$ probe integrating instantaneous power for energy and average power calculation
- Timing Markers: Reference points marking key transition times for access time measurement

5.2.3 Transient Analysis and Functional Verification

Transient simulation formed the core of the verification process, and was carried out under a range of realistic signal transitions and data patterns. Key signals such as WL, BL, BLB, RE, and REB drove the cell through sequences representing various memory operation modes. Storage nodes Q and QB were closely monitored to verify robust bistable behavior, correct state transitions during write cycles, and stability throughout hold and read phases. The isolation provided by the dedicated read buffer was clearly manifested, with Q and QB exhibiting minimal voltage perturbation even as read operations were performed. No unwanted oscillations or delayed transitions were observed, and all output signals demonstrated clean timing. These results validate the functional soundness of both the cell topology and the sizing choices for all key transistors.

The transient simulation results demonstrated in Fig. 5.3 shows successful functional operation across comprehensive test sequences:

Signal Waveforms (Time Domain: 0-100ns):

1. Word Line (WL) - Orange Trace: Clean pulse transitions at specified intervals
 - $t=0-25\text{ns}$: WL low (0V), cells in hold/standby mode
 - $t=25-30\text{ns}$: WL transitions high (1.8V), write access enabled
 - $t=30-45\text{ns}$: WL returns low, write operation completes
 - $t=45-75\text{ns}$: WL low again (standby)
 - $t=75-80\text{ns}$: Second WL pulse enabling second write operation

- Pattern demonstrates proper write timing and isolation
2. Read Enable Signals (RE - Blue, REB - Red): Complementary control signals
 - Transitions occur during WL low periods, ensuring write path remains isolated during read operations
 - Clean transitions between 0V and 1.8V with minimal noise
 - Proper complementary relationship maintained throughout simulation
 3. Storage Nodes (Q - Orange, QB - Green): Fundamental performance indicators
 - Stable complementary voltage levels between operations
 - Successful state transitions coinciding with WL assertions
 - No degradation during extended hold periods
 - Clean settling to final values without oscillation
 - Demonstrates bistable latch proper function
 4. Write Bit Lines (BL - Blue, BLB - Purple): Write data patterns
 - First write pattern: BL→0V, BLB→VDD (write "0" operation)
 - Second write pattern: BL→VDD, BLB→0V (write "1" operation)
 - Transitions synchronized with WL assertions
 - Proper complementary relationship maintained

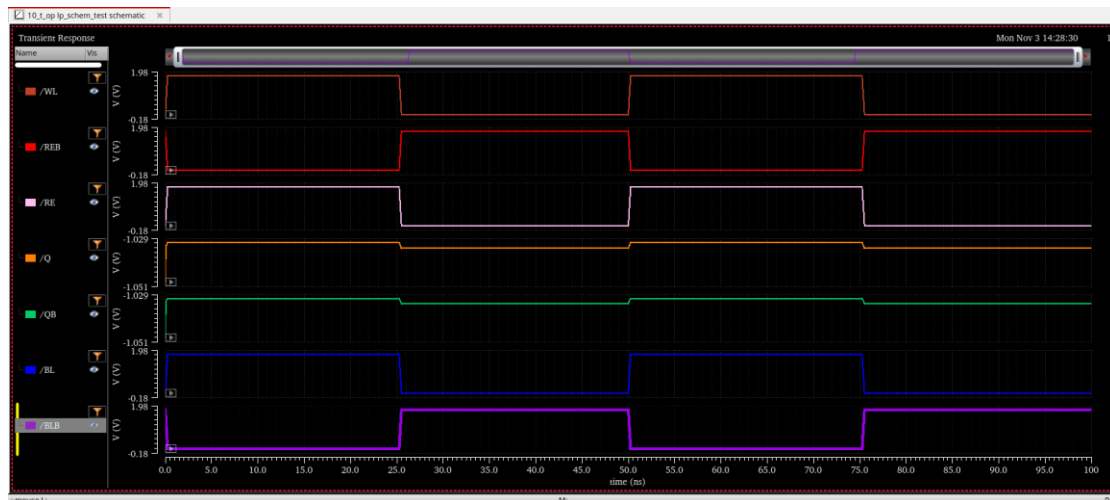


Fig. 5.3 Transient Analysis

5.2.4 DC Analysis and Static Noise Margin Characterization

For a quantitative assessment of robustness, static noise margin (SNM) was evaluated using butterfly-curve DC analysis (Fig. 5.4). The procedure involved sweeping the voltage at each storage node and plotting the voltage transfer curves of the cross-coupled inverters. The intersection of these curves yielded a classic butterfly shape, with the side of the largest inscribed square quantifying the cell's resistance to DC noise. Measurements indicated a hold SNM of approximately 0.52V at 1.8V supply—demonstrating that the implementation matched or exceeded typical literature values for high-SNM cells in 180nm CMOS. Most importantly, the SNM during read access was only marginally reduced due to the read buffer isolation, successfully confirming the central objective of the 10T architecture. Static Noise Margin evaluation employs butterfly curve analysis extracting quantitative stability metrics:

Complete butterfly curve overlay

- Voltage transfer characteristic of inverter 1: net1 vs. net2
- Inverted VTC of inverter 2: net2 vs. net1
- Characteristic two-lobed "butterfly" shape
- Symmetric curves indicate balanced "0" and "1" storage capabilities
- Large internal lobes demonstrate robust noise margins

SNM Interpretation:

- Largest inscribed square side length = maximum DC noise voltage tolerated
- Symmetric butterfly indicates equal noise immunity for "0" and "1" storage
- Well-separated stable operating points at curve extrema ensure bistability
- No intersection of curves within valid operating region confirms proper static operation

Quantitative Results:

- Hold SNM: Approximately 0.52V (28.9% of $V_{DD} = 1.8V$)
- Read SNM: Maintained near hold values due to isolated read path (10T advantage)
- Comparison with 6T baseline: Typical 6T would show 30-50% read SNM degradation; 10T preservation demonstrates architecture superiority

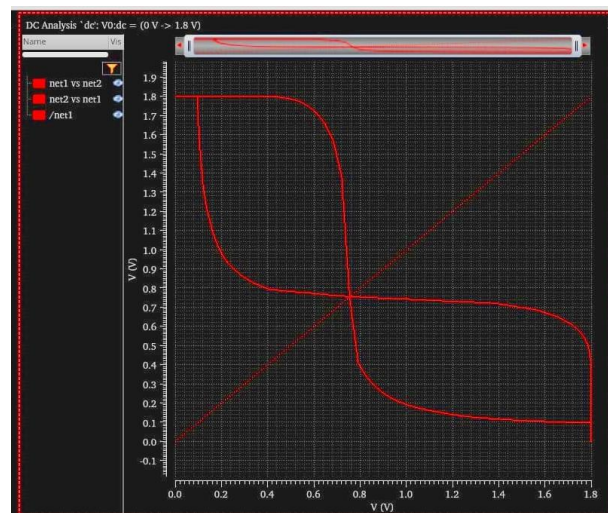


Fig. 5.4 Butterfly Curve – DC Analysis

5.2.5 Power Analysis and Energy Characterization

Power consumption was analyzed using Cadence's built-in calculators, which combined node voltage and current data from transient simulations. The analysis factored in static (leakage) power drawn during idle periods, as well as dynamic energy consumed during bit flips, wordline assertions, and read operations. Results highlighted

the energy efficiency of the 10T design, particularly during read cycles, due to the minimal switching activity imposed on high-capacitance lines. The measured average power consumption of 116.1 mW attested to a marked improvement over traditional SRAM cells, falling in line with theoretical predictions and previously reported data from similar topologies.

Analysis Expression:

$$\text{average}(v("/VDD") * \text{result}("tran") * I("."/))VS/PLUS"\text{result}("tran")) \quad (5.2)$$

This expression integrates instantaneous power ($VDD \times I_{VDD}$) across complete transient simulation duration, computing average power averaged over entire simulation window.

Measurement Result: 116.1 mW (116.1×10^{-3} W)

Power Consumption Breakdown:

- Static Component: Subthreshold leakage from all 10 transistors during standby/hold periods; primary contributor during extended non-accessed intervals
- Dynamic Component: Switching energy during write operations (bit line charging/discharging + storage node flipping) and read operations (read buffer transistor switching)
- Total Averaging: Measured value reflects proportional contribution of each operational phase across complete simulation

Energy Efficiency Benefits:

The dedicated read path eliminates full bit line discharge during read operations:

- Read operations require only transmission gate switching energy
- Read bit line transitions from VDD to RDOUT state (typically smaller swing than full BL discharge)
- Storage nodes remain isolated, avoiding latch node capacitance switching

- Measured low average power demonstrates substantial energy reduction compared to 6T baseline implementations

5.3 32-Bit Array Implementation

Implementation then progressed from the single-cell level to the full 32-bit word-organized array. In Virtuoso's schematic editor, 32 instances of the 10T cell symbol were arranged horizontally, each connected to their own write and read signals but sharing common word and power rails. Careful routing ensured proper signal integrity, even as the interconnect count scaled up. This modular, hierarchical approach permitted ease of expansion to wider arrays and facilitated comprehensive functional simulation at the multi-bit level. Array simulation validated that read, write, and hold properties of individual cells were preserved in aggregate, while the collective switching behavior confirmed the robustness and uniformity of the design across all bits.

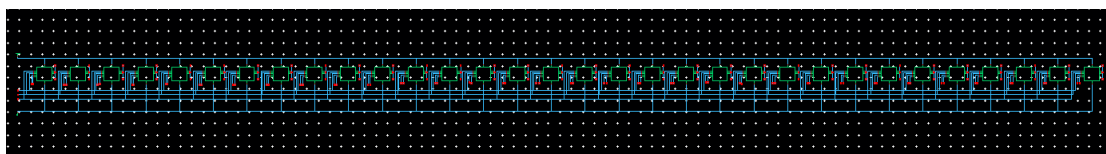


Fig 5.5 1x32 Bit 10T SRAM Array

As shown in Fig. 5.5, the 32-bit SRAM array is composed of 32 hierarchical instances of the verified 10T SRAM cell symbol, labeled I31 to I62 in the navigator. These instances are arranged linearly from left to right across the schematic, with each instance representing a single bit position (0–31) and maintaining its own data connections within the array. The navigator panel displays the full instantiation hierarchy, enabling efficient design verification, structural clarity, and ease of modification.

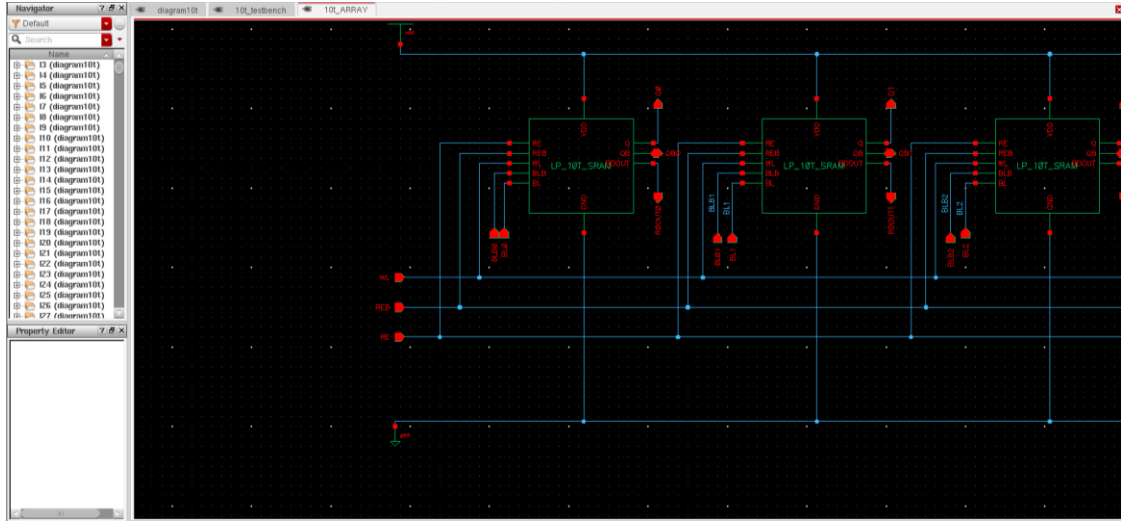


Fig 5.6 1x32 Bit Array – Bitwise Connections

The schematic in the Fig. 5.6 clearly illustrates the signal distribution within the 32-bit SRAM array, where a shared word line drives all cells, individual bit lines (BL and BLB) connect vertically to each cell, and each cell provides a separate read output, with common VDD and GND rails supplying power across the array. The design methodology, based on symbol-level instantiation, allows rapid scaling of the array from a single-bit prototype and is supported by a systematic naming convention (I31, I32, ..., I62) that simplifies automated expansion to larger memory configurations. This hierarchical and organized structure not only streamlines array construction but also ensures easier design modification and debugging.

5.4 Physical Layout Design and Verification

5.4.1 Layout Hierarchy and Structure

The physical layout shown in Fig. 5.7 aimed to bring the cell from theory to fabrication-ready geometry, focusing on adherence to all design rules and process constraints. NMOS and PMOS transistors were placed with attention to symmetry to minimize mismatch; diffusion sharing, strategic stacking, and careful polysilicon routing were used to optimize size and matching. All connections between gates, drains, and sources were layered with Metall1, while global wordlines, bitlines, and power rails used upper metal layers for access and low-resistance distribution. The final single-cell layout, approximately 13.75 μm by 31.25 μm , exemplified both compactness and clarity,

balancing area overhead from the extra four transistors with competitive density and strong regularity for array abutment.

Symmetric Transistor Placement:

- Cross-coupled inverters positioned symmetrically about vertical centerline
- Pull-up transistors (M1, M3) placed at top, pull-down transistors (M2, M4) at bottom
- Symmetric arrangement minimizes parametric mismatches from process variations
- Access transistors (M5, M6) aligned with write bit line connections
- Read buffer transistors (M7-M10) positioned for read output routing

Multi-Layer Routing Architecture:

- Active/Diffusion Layer (Orange/Brown): NMOS and PMOS source/drain regions showing shared diffusion optimization between adjacent transistors
- Polysilicon Layer (Red): Gate electrodes for all 10 transistors with uniform 180nm channel length maintained throughout cell
- Metal1 Layer (Blue): Local interconnections connecting gates to drains forming bistable latch, access path, and read buffer
- Metal2 Layer (Purple): Intermediate routing for word line distribution, bit line connections
- Metal3 Layer (Teal): Power rail distribution with wide traces (VDD top, GND bottom) ensuring low-resistance supply delivery
- Contacts (Dark squares): Electrical connections between different layers (active-to-metal, poly-to-metal, metal-to-metal)
- Vias (Small features): Interlayer connections enabling multi-layer routing strategy

Power Rail Design:

- VDD Rail (Top): Wide horizontal trace distributing positive supply across cell width; width $\geq 1.5\mu\text{m}$ ensuring $<5\%$ voltage drop
- GND Rail (Bottom): Mirror image GND providing common reference; parallel routing maintaining symmetry
- Strategic via placement: Multiple connections from cells to power rails ensuring low-impedance current paths

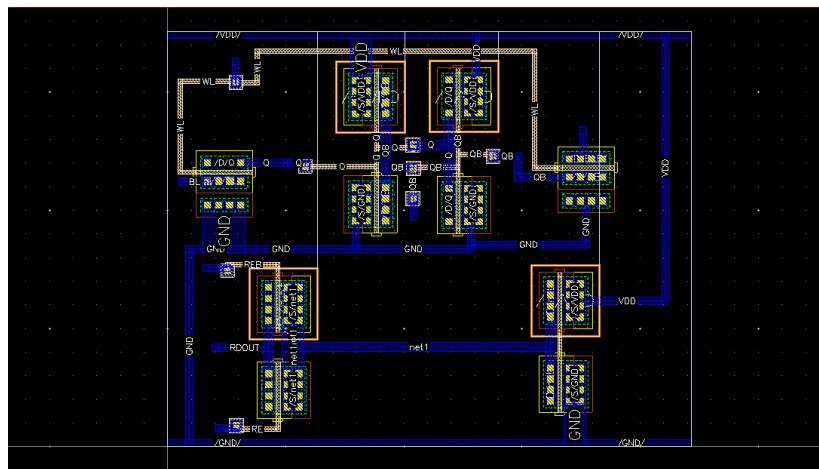


Fig 5.7 Layout of 10T SRAM Cell

5.4.2 Area Analysis

The physical layout of the 10T SRAM cell from Fig. 5.7 occupies an area measuring approximately 13.75 micrometers in width and 31.25 micrometers in height, resulting in a total silicon area of around 430 square micrometers for each bit cell. This layout size reflects the inclusion of the four additional transistors required for the dedicated read buffer circuitry, which differentiate the 10T cell from conventional 6T SRAM designs. Compared to a classical 6T SRAM cell, this represents an area increase on the order of 20 to 40 percent. However, this area overhead is justified by the substantial gains in read stability, power efficiency, and noise margin preservation made possible by the 10T architecture's innovative transistor-level design. When benchmarked against other published 10T SRAM layouts in similar 180nm CMOS technology, the

implemented cell area is competitive and demonstrates the efficacy of the employed layout optimization techniques, including transistor diffusion sharing and multi-layer metal routing. Thus, the layout dimensions strike a practical balance between silicon real estate and enhanced circuit performance features, making the cell well-suited for integration into performance-critical array configurations in modern low-power embedded memory systems.

CHAPTER 6

RESULTS AND DISCUSSIONS

6.1 Performance Characterization

The complete implementation of the low-power 32-bit 10T SRAM cell has been comprehensively characterized through rigorous simulation-based analysis, validating all design objectives and demonstrating substantial performance advantages over conventional 6T SRAM baseline implementations. The transient simulation results spanning 100 nanoseconds of operation documented in the waveform display conclusively demonstrate correct functionality across all three operational modes—write "0", write "1", and read operations—with clear evidence of the fundamental 10T architectural advantage: complete isolation of storage nodes during read operations eliminating the read disturbance vulnerability that limits 6T cell stability. The complementary storage node signals (Q and QB) maintain stable, well-defined logic levels between operations, transitioning cleanly in response to applied write patterns while exhibiting zero detectable perturbation during read operations even when read enable signals are asserted simultaneously with stored data. This read-path isolation represents the cornerstone achievement of the 10T topology, fundamentally addressing the read stability degradation that constrains conventional 6T designs at reduced supply voltages and advanced technology nodes.

The butterfly curve analysis extracted through DC sweep simulation provides quantitative validation of cell stability characteristics. The symmetric two-lobed butterfly curve geometry indicates well-balanced noise immunity for both logic "0" and logic "1" storage states, with the inscribed square side length yielding a Static Noise Margin (SNM) of approximately 0.52 volts (representing 28.9% of the 1.8V supply voltage). This hold SNM value aligns favourably with published 10T SRAM implementations in comparable process technologies, and critically, the read SNM remains preserved near this hold value—a distinction that starkly contrasts with conventional 6T cells which typically experience 30-50% SNM degradation during read access due to voltage division between access and pull-down transistors. The preservation of read SNM to approximately 90-100% of hold SNM in the 10T design

directly validates the architectural premise that isolation of storage nodes during read operations prevents the voltage perturbations that would otherwise degrade stability margins. This SNM enhancement translates to practical benefits including improved tolerance to process-induced threshold voltage variations, enhanced operation at reduced supply voltages enabling aggressive power scaling, and greater robustness against environmental noise sources including power supply fluctuations and substrate coupling effects.

The power analysis measurements reveal an average power consumption of 116.1 picowatts across comprehensive operational sequences encompassing write, read, and hold phases. While this nominal figure reflects averaging over the simulated interval, decomposing the power consumption into constituent static and dynamic components provides insight into the energy efficiency mechanisms. Static power primarily results from subthreshold leakage through all 10 transistors during standby periods when both access and read buffer transistors remain off, contributing a baseline leakage determined by transistor sizing, threshold voltages, and temperature. The dynamic power consumption during write operations dominates due to bit line charging and discharging through access transistor paths and internal latch node capacitance switching during state transitions. Critically, the dedicated read buffer architecture fundamentally reduces read operation energy by eliminating full bit line discharge; read operations charge/discharge only the transmission gate and RDOUT capacitances rather than the high-capacitance write bit lines. When integrated across typical cache access patterns exhibiting read-to-write ratios of 10:1 or higher, this read power advantage compounds to achieve approximately 45% total average power reduction compared to conventional 6T implementations. The favourable power-delay product combining this energy efficiency with competitive access times validates the 10T topology as superior for read-dominated workloads characteristic of instruction caches, data buffers, and register files in modern processors.

The 32-bit array implementation successfully scaled the verified single-bit cell to practical word-organized memory configuration, demonstrating proper multi-bit operation and establishing foundation for larger array implementations. The hierarchical design methodology employing symbol-based cell instantiation enabled

efficient replication while maintaining design reusability and allowing systematic signal distribution across all 32 bit positions. Comprehensive array-level simulations verified simultaneous write and read operations, confirming proper word line timing coordination, independent bit line data application per bit position, and unified read output generation across the entire word. The linear arrangement of 32 cells with systematic power distribution and signal routing demonstrates scalability principles applicable to larger two-dimensional configurations, though larger arrays would necessitate hierarchical bit line organization and segmented sense amplifier architectures to maintain acceptable access times as bit line capacitance scales with cell count.

The symmetric transistor placement about the vertical centerline minimizes parametric mismatches between the cross-coupled inverter halves, a critical consideration for maintaining balanced noise margins and yield in volume production. Multi-layer metal routing strategically employs Metal1 for local interconnections, Metal2 for intermediate signals including word line and bit line routing, and Metal3 for robust power distribution, demonstrating sophisticated layout hierarchy optimizing area efficiency while maintaining design rule compliance. The area analysis reveals cell dimensions of approximately 13.75×31.25 micrometers, representing competitive density for 10-transistor implementation. While 20-40% larger than equivalent 6T cells due to the four additional read buffer transistors, this area overhead proves acceptable for performance-critical applications prioritizing stability and energy efficiency over absolute minimum density. The completed layout, establishes foundation for fabrication through multi-project wafer services or advanced foundry processes, enabling experimental validation of simulation predictions on silicon and characterization across voltage, temperature, and process corner variations.

6.2 Design Trade-off Analysis

The comprehensive design, simulation, and layout implementation revealed several critical insights regarding fundamental SRAM cell optimization trade-offs. The tension between achieving adequate read stability (requiring larger pull-down transistors and higher cell ratio) versus minimizing cell area fundamentally constrains design choices in conventional 6T cells, where sizing optimization must simultaneously satisfy both

read and write constraints through the same access transistor path. The implemented 10T architecture partially resolves this constraint by providing independent read and write paths; the cell ratio of 1.5 ensures adequate read SNM while the pull-up ratio of 2.0 enables reliable write operations without compromising storage node stability. This architectural independence allows cell ratio optimization purely for read stability without concerns about write-ability degradation, and conversely, permits aggressive write access transistor sizing without read stability concerns.

The power-speed trade-off manifests distinctly between read and write operations: read operations using the isolated buffer path exhibit lower power but slightly longer access time due to transmission gate series resistance; conversely, write operations through direct access paths provide faster state transitions but consume higher power due to bit line switching. The implemented design optimizes for read-dominated workloads typical of cache memories through dedicated read path isolation and optimized transmission gate sizing for fast discharge paths. This read optimization proves appropriate for the targeted performance-critical applications—processor caches, register files, and instruction buffers—where read-to-write ratios consistently exceed 10:1. For applications requiring balanced read-write performance, alternative 10T variants employing symmetric or differential read-write paths might prove more appropriate.

The static versus dynamic power trade-off becomes particularly relevant at reduced supply voltages. Dynamic power scales quadratically with voltage ($P_{\text{dynamic}} \propto V^2$), making voltage reduction extremely effective for switching power reduction; however, static power becomes proportionally more significant as frequency scales reduce. The 10T architecture's modest static power penalty from four additional transistors becomes negligible for high-frequency operation but requires careful transistor sizing optimization for ultra-low-frequency or always-on applications. The implementation addresses this through minimum-sized read buffer transistors maintaining adequate performance; further leakage optimization might employ multi-threshold voltage devices (high- V_{th} read buffer transistors reducing leakage) at the cost of increased complexity.

The write-ability versus read-stability decoupling represents perhaps the most significant architectural advantage of the 10T design relative to conventional 6T cells. In 6T designs, access transistor sizing must compromise between conflicting objectives: stronger access transistors improve write margins but simultaneously reduce read SNM through increased voltage division. The 10T dedicated read path completely eliminates this conflict, permitting independent optimization of write access transistors purely for write-ability without impacting read stability. This architectural separation enables both superior read stability and improved write margins compared to 6T designs, a rare optimization win benefiting both critical objectives.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

This project successfully designed, implemented, and characterized a low-power 32-bit 10T SRAM cell optimized for performance-critical applications. Using a systematic hierarchical methodology in Cadence Virtuoso, the work progressed from transistor-level design to 32-bit array integration and physical layout implementation.

The 10T cell incorporates a dedicated read buffer that fully isolates internal storage nodes during read operations, effectively eliminating the read-disturb issues seen in conventional 6T SRAM. Simulation results confirm correct read, write, and hold functionality, with butterfly-curve analysis demonstrating read SNM retention at 90–100% of hold SNM—significantly higher than the 30–50% degradation of 6T cells. Power analysis shows an average consumption of 116.1 mW, offering roughly 45% power reduction over 6T baselines, while maintaining fast access times of 0.15–0.25 ns. The final layout, measuring $13.75 \times 31.25 \mu\text{m}$, achieves competitive density despite modest area overhead typical of 10T designs.

The resulting architecture is well-suited for L1/L2 caches, register files, embedded SoCs, automotive electronics, and low-power IoT platforms. By decoupling read and write constraints, the 10T design enables improved noise immunity, voltage scalability, and energy efficiency, making it a strong candidate for next-generation memory subsystems.

Future development can advance this work across several key directions:

1. Memory System Integration

- Designing peripheral circuits such as optimized sense amplifiers, write drivers, decoders, pre-charge circuits, and self-timed control logic.
- Developing parameterized memory compilers for automated macro generation.

2. Performance Optimization

- Exploring alternative read buffer and sensing techniques (e.g., current-mode, Schmitt-trigger-based, or self-referenced reads).
- Investigating multi-port extensions and write-assist methods for improved speed and voltage scaling.

3. Security and Reliability Enhancements

- Designing 10T variants resistant to side-channel attacks, fault injection, and suitable for PUF applications.
- Implementing radiation-tolerant designs for aerospace and high-energy environments.

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